Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2418)

Key

SECTION – I

1. B
2. A
3. C
4. C
5. D
6. C
7. B
8. D
9. B
10. C
11. A
12. D
13. B
14. D
15. C
16. B
17. 3241
18. A
19. D
20. 3412
21. B
22. 3241
23. C
24. C

SECTION – II

1. C
2. B
3. C
4. D
5. A
6. A
7. A
8. D
9. 8
10. 122
11. A
12. 10
13. 30
14. B
15. 65
16. C
17. C
18. B
19. B
20. D

SECTION – III

1. 11
2. 16
3. D
4. 3
5. D
6. C
7. A
8. 28
9. B
10. B
11. A
12. C
13. D
14. 4
15. A
16. C
17. C
18. 144000
19. D
20. 20
21. A
22. B

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 459

1. Option A: Historical thought for the Renaissance historian ***had man at the centre***, compared to previous historical thought, and not the other way around. So this option is easy to eliminate. Option A is not the answer.
Option B: This can clearly be understood from: '*Man, for the Renaissance historian, was not man as depicted by ancient philosophy, controlling his actions and creating his destiny by the work of his intellect, but man as depicted by Christian thought, a creature of passion and impulse.*' So, the human of Christian thought is one of passions while the human of ancient philosophy was creating a destiny using his intellect. Option B is the answer.
Option C: The Renaissance historian took human history as a function of humans depicted by the Christian thought. Man as depicted by Christian thought is not defined by Renaissance conception. The para doesn't really indicate anything as a particular Renaissance conception. Option C is not the answer.
Option D: While the inclinations of this option are correct, the para doesn't really establish that history is a ***consequence*** of passion or intellect. It merely points out that one is the history of passions whereas the other is the history of man creating his destiny. Hence, Option D can be eliminated. Choice (B)

2. Option A: The author clearly points out that the basis for determining the general plan of history a priory – the great theological and philosophical systems had ceased to command assent. The change was depicted as human actions being held important (unlike earlier where it was considered a divine plan). So, we can infer from both the statements that the a priori determination of the general form of history looked at events as being part of a divine plan. Option A is the answer.

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Option B: Accurate scholarship was important because it allowed historians to pay the importance to humanistic view of events, not to proscribe (forbid or exclude) the importance of human actions. So, this option contradicts the information in the para. Option B is not the answer.
Option C: Renaissance ideas depicted a humanistic view of actions as opposed to the divine plan. So, theological systems ceased to command assent (agreement). The para doesn't talk about the 'ascension' or rise of humans anywhere. So, the question of a path of human ascension doesn't arise. Option C is not the answer.
Option D: Theological systems depicted events as being part of a divine plan. This can be inferred from the para. We cannot really say that theological systems 'distorted' the actual events. Yes, events didn't matter because of the a priori determination of a plan of history. But actual history cannot be said to have been distorted. Option D is not the answer. Choice (A)

3. Option A: The author doesn't really say that there are two parts to history: what we can remember and what we cannot. In fact, the author points out that history is not really needed when we can remember. So, Option A is not the answer.
Option B: The author doesn't bring in the concept of interpretation. Yes, historical investigation was discussed, but interpretation of events as they happened is not what is wrong with Bacon's idea of history. Option B is not the answer.
Option C: This is what the author was trying to suggest when he pointed out that '*Bacon's definition of history as the realm of memory was wrong,* ***because the past only requires historical investigation so far as it is not and cannot be remembered.***' So, history is required because of what cannot be remembered and hence, is out of the domain of memory. This perfectly invalidates Bacon's idea of history being a function of our memory of actual events as they happened. Option C is the answer.
Option D: If this is not the primary function, it could still be a secondary function. The author doesn't invalidate by saying something else is more important. The invalidation comes in the form of outright reversal of Bacon's idea of history – that historical investigation is not about memory but about that which cannot be remembered. So, Option D is not the answer. Choice (C)

4. Option A: The author wasn't really building a hypothesis that Bacon's contemporaries didn't agree with his idea. Such a discussion has not been approached in the passage. The author's hypothesis was more in the lines of why Bacon was wrong. Option A is not the answer.
Option B: While it is close, the focus was not on the idea that history can be rebuilt without memory. The focus was on how history can be rebuilt from data (more about how it can be rebuilt, rather than what it can be built without).
Option C: The author is trying to build this hypothesis that a historian's task is not to put down what can be remembered but to investigate what isn't remembered. So, Camden's example shows it can be done, that it is already being done, even as Bacon wrongly believed in memory's domain in history. So, Option C is the answer.
Option D: The author mentions that Bacon never posed this question. The author himself doesn't answer the question of how the historian's understanding works to supplement the deficiencies of the memory. So, this cannot be the hypothesis, that the understanding supplements the deficiencies in memory (Camden was using data, not understanding). Choice (C)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 540

5. The question that author poses is "What is changing at a fundamental level, when the surface style changes?" The Semperians' answer to this question is "purpose, material and techniques". But the author feels that his "question only has the appearance of having been answered".
Option A: The Semperians' consider purpose, material and technique to be determinants of style. The author is of the opinion that these three factors are not the determinants of style and style can remain unchanged even with changes in the above three factors. This option is incorrect because it mentions that Semperians assumed that style can remain unchanged with changes in purpose, material and technique which implies that these three factors cannot be the determinants of style.
Option B: While the author does mention that ignoring intellectual factor is a mistake, this is not the reason that the author calls their answer an apparent answer. We can infer from the passage that including an intellectual factor in the Semperians' answer will not answer the author's question because he considers that purpose, material and technique are not the determinants of style (with or without the inclusion of an intellectual factor). Hence, this is not the correct answer.
Option C: The reason why the author believes that his "question only has the appearance of having been answered" is because "style means something quite different from what it meant in our question". The author does not call their answer an "apparent answer" because they did not consider the influence of *Kunstwollen*.
Option D: The author mentions that "according to this answer, style means something quite different from what it meant in our question, and that our question only has the appearance of having been answered". Hence, the concept of style according to Semperians is different and the answer that they provided is in line with the concept of style according to their definition. But it is not the same as the concept of style according to the author. Hence, this is the correct answer. Choice (D)

6. The author talks about the relation between art drive and purpose, material and technique in the penultimate paragraph of the passage.
Option A: The author mentions that "purpose, the material and the technique also change, but they are negative factors, mere "frictional coefficients," which have to be subtracted in order to recognize the pure "direction of the art drive"". Further, he also mentions that two of them, material and technique, are "partly dependent on the direction of the art drive". Hence, this option is incorrect as it states the vice versa.
Option B: The author mentions that two of them depend on art drive. He also calls the direction of art drive as "the positive determining factor". Hence, this is not the correct answer.
Option C: The author mentions that the influence of purpose, material and technique must be "subtracted" to recognize the pure "direction of the art drive". We can infer from this that the influence of purpose, material and technique are more of an interference which obscure the direction of the art drive. Hence, this is the correct answer.
Option D: The author mentions that material and technique are "**partly dependent** on the direction of the art drive". He does not mention that purpose is dependent on the direction of the art drive. Even for the other two variables, we cannot say that they are dependent on art drive (since the author mentions that they are only partly dependent). Hence, this is not the correct answer.
Therefore, the correct answer is option C.
Choice (C)

7. The author talks about Reigl in the penultimate paragraph of the passage.
Option A: Further, in the fourth paragraph of the passage, the author asks "We know the dependent variable factor, namely the style of works of art; what is the variable independent of all others?". According to Reigl, the independent variable is the direction of *Kunstwollen*. Hence, we can infer that this independent variable is independent of the "style of works of art". Therefore, Reigl would most probably not agree

8. In the third paragraph of the passage, the author opines that the "concept of style as something grasped intuitively and described through highlighting individual stylistic characteristics is by its very nature extremely erratic and uncertain".
Option A: The author mentions that "The purely empirical depiction of styles by their individual characteristics is not scientific". Using empirical depiction results in "extremely erratic and uncertain" definition of concept of style. Hence, defining it in a scientific manner will not result in an erratic and uncertain definition. Therefore, this is not the correct answer.
Option B: The author mentions that "The purely empirical depiction of styles by their individual characteristics is not scientific, in the proper sense of the word. ***It stops with merely outward description***." However, he does not mention that empirical depiction of styles, which can be inferred to mean forms, should not be used to define the concept of style. We can infer that this alone is not sufficient because it "stops with merely outward description." Hence, this is not the correct answer.
Option C: The author does not refer to the direction of art drive in this paragraph. Hence, this is not the correct answer.
Option D: The empirical depiction (i.e., outward appearance) is not scientific. This definition will be erratic and uncertain. It stops with merely the outward description. Therefore, this will not result in a comprehensive definition of the concept of style. Hence, the correct answer is option D.
Choice (D)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 503

9. Option A: While this may be true, it is different from the paradox that has been mentioned in the question. The paradox applies only to the diversity approach, and the contradiction inherent in it is how employees have to be treated the same way and yet have needs that need to be dealt with differentially. Option A is not the answer.

with this statement.
Option B: According to Reigl, *Kunstwollen* is the independent variable. This is independent of the style of works of art. Hence, Reigl will most probably agree with this statement.
Option C: From the penultimate paragraph, we can infer that the direction of *Kunstwollen* drives these changes in forms and styles but we cannot infer whether this will help identify forms across space and time. Hence, this is not the correct answer.
Option D: In the third paragraph, the author mentions that the concept of style "described through highlighting individual stylistic characteristics is by its very nature extremely erratic and uncertain". However, the author does not talk about *Kunstwollen* in this context. Hence, this is not the correct answer.
Therefore, the correct answer is option B.
Choice (B)

Option B: This represents the paradox aptly. Individual needs of employees are different and hence, managers need to treat them differently. However, employees need to be treated the same way. That is the sameness-difference paradox inherent in the diversity approach. Option B is the answer.
Option C: While this is an observation made by the author in a previous segment, it doesn't reflect any apparent contradiction. It doesn't really involve a paradox either, a mere judgment of how diversity fails sometimes to achieve the right results. Hence, Option C is not the answer.
Option D: The diversity approach may benefit ethnic minority employees, but the comparison that it benefits them more than it benefits others is misplaced. Also, there is no paradox in this observation/opinion. Option D is not the answer.
Choice (B)

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10. Option A: This has been mentioned as a disadvantage of diversity approach. The passage clearly indicates that diversity discourses do not confront power relations, or existing structural issues. They do benefit ethnic minorities to a certain extent but do not provide a solution to the root cause as understood from, '*They do not tackle the deep, structural problems discussed by other commentators*.' Option A is not the answer.
Option B: This has also been mentioned in the sentence: '*Diversity discourses are not constructed to confront power relations, dominant ideologies or organizational goals*.' So, whatever issues exist at the organisational level, continue to exist despite the diversity discourse. Option B is not the answer.
Option C: This is not a downside of the diversity approach because according to the passage, '*Control of policies and practices has always resided with management*,' and hence, it isn't something that can be attributed to the diversity discourse. Option C is the answer.
Option D: This is a downside as mentioned in the passage. A lot of managers resorted to ethnic stereotypes as they didn't quite understand the sensitivity of diversity approach, as explained in '*managers they interviewed resorted to stereotypes to explain differences and were interested only in how such differences could be deployed in relation to organizational goals*.' Option D is not the answer.
Choice (C)

(In other words, the example was used to explain, this is how this approach will benefit the organisation, NOT, this will still not solve this problem.) Option A is not the answer.
Option B: This contradicts the research because it agrees with the managers' viewpoint. Managers try to use stereotypes to explain differences amongst employees, thus getting the complete focus of a diversity discourse wrong. Option B is not the answer.
Option C: The concern that they had was with respect to how diversity is actually misused. While it is true, that the definition of diversity is attached to stereotypes, the point of their research wasn't to show this. They had more things to conclude than just stop at this basic observation. Option C is not the answer.
Option D: The differences could be explained as inadequacies or as vulnerabilities, allowing the organisation to benefit from them one way or the other. That is what the illustration showed us – that diversity under the discretion of managers will be reduced to what benefits or doesn't benefit the organisation. Option D is the answer.
Choice (D)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 512

11. Option A: The author's main concern in the passage is that the diversity discourse is flawed because it doesn't address the structural problems and makes diversity a business move, rather than a social enhancement. So, the statement that it is a start, and will lead to better things, and eventual goals, despite its imperfect implementation alleviates (or reduces the negativity in) the concerns of the author. Option A is the answer.
Option B: This is already true according to the author and the concern held by the author stays regardless of this minor benefit. Hence, Option B is not the answer.
Option C: While this points to a potential benefit of the diversity discourse – benefiting the ethnic minorities in a way equal opportunities approach doesn't – it doesn't touch/address the concerns of the author in the passage, about the flaws in the diversity discourse (maintaining existing power relations). Option C is not the answer.
Option D: We are still not sure, from the passage, if making managers responsible is a good thing or bad. Hence, Option D is not the answer.
Choice (A)

12. Option A: This is a concern presented by the author, but it isn't the theory presented by Zanoni and Janssens. It is an extrapolation of their conclusion that diversity discourses reflect existing power relations in an organisation. Moreover, the example was to show how the discourse is used to fit the organisational context and benefits

13. Refer to paras 1, 2 and 4.
Faust is a scholar who is highly successful yet dissatisfied with his life, which leads him to make a pact with the Devil, exchanging his soul for unlimited knowledge and worldly pleasures. "Faust" and the adjective "Faustian" imply a situation in which an ambitious person surrenders moral integrity in order to achieve power and success for a delimited term.
Option A: Choice A is true but it is incomplete as an answer to the question. No period has been more penetrated by science, nor more dependent upon it, than the past half century.
Option B: Refer to para 1. There are few things that have more changed our world than has science. Scientists and their discoveries have helped transform material conditions and opened up new social and moral vistas. Yet it is the very notion of human-directed change that many people today find so troubling. No period has been more penetrated by science, nor more dependent upon it, than the past half century. Yet no period has been more uneasy about it, nor felt more that the relationship with scientific knowledge is a Faustian pact. ...Refer to para 4. ***What many people fear is a science that disturbs their moral compass.*** Hence choice B is the answer.
Option C: There are few things that have more changed our world than has science. Scientists and their discoveries have helped transform material conditions and opened up new social and moral vistas. Yet it is the very notion of human-

directed change that many people today find so troubling. ...No science has seemed more to call all in doubt than the science of biology. Choice C is true but it is not the reason for the question.
Option D: Choice D is true but is not the specific answer to the question. Choice (B)

14. Option A: Choice A is true but it is limited to the last para. Each generation must be allowed to struggle with human nature as it is given to them, and not with the irreversible biological results of their forbears' actions. Choice A is a close choice but it is not the primary concern of the author.
Option B: Choice B is out of scope. The author does not justify the need to examine scientific discoveries under the microscope of reality.
Option C: Opinion formers in society worry that man is now playing God, remaking nature in his own image. Choice C is true but is not the primary concern of the author.

Solution for question 17:

17. On a careful reading of the sentences it can be observed that sentence 3 is a general sentence that begins the paragraph. It introduces the term "feud" and the background "family fights another". Sentences 3 and 2 form a mandatory pair. Sentence 2 extends the idea mentioned in sentence 3 – "one family fights with another" in sentence 3 links with "When lots of families fight with one another ..... mountain range" in sentence 2. Sentence 4 follows sentence 2. "the Appalachian pattern" in sentence 4 points to "a pattern" in sentence 2. Sentence 4 (What was the cause?) is linked with sentence 1 (The consensus reached – this addresses the question asked in sentence 4). So, sentence 1 follows sentence 4. "that region" in sentence 1 points to "Appalachian" in sentence 4. So, 3241.
Ans: (3241)

Option D: Choice D is true as the first four paras show. People are enamoured by the progress in science and the power it gives, on the one hand, and the fear, the consequences and moral dilemma, it poses on the other. Choice (D)

15. What many people fear is a science that disturbs their moral compass, upsetting traditional ideas of Man and nature, a science that promises new forms of control over nature, new types of mastery over human destiny. So choices A, B and D are true and are not the answers.

What many people are drawn to is a science *that provides science and comfort,* ***that turns an explanation about the human condition into a parable about fate***. Choice (C) is the answer. Choice (C)

16. Option A: Mary Douglas and Aaron Wildavsky observe, 'nature plays the role of general arbiter of human designs more plausibly than God'. Today, nature is rapidly turning into a new deity to whom we turn for moral answer and personal comfort. So choice A is not definitely false.

Option B: Choice B is definitely false. 'Have we the right', the molecular biologist Ervin Chargaff asks, 'to counteract, irreversibly, the evolutionary wisdom of millions of years? Each generation must be allowed to struggle with human nature as it is given to them, and ***not with the irreversible biological results of their forbears' actions***. Choice B is definitely false and the answer to the question.

Option C: As Norman Levitt put it 'The "natural" is the virtuous opposite of the degraded manifestations of humanity's fallen state.' Choice C is definitely true. It is not the answer.

Option D: Bryan Appleyard is terrified by the way that science has invaded the human realm. "The new biology entails the thwarting of nature at a very fundamental level. Genetics must be contained, humbled." Choice D is correct and is not the answer. Choice (B)

Solution for question 18:

18. Option A: The author of the paragraph first makes a positive statement about translators and then goes on to qualify this statement. He also cites important limitations and the nature of those limitations (.....alter the rhythm, the syntax and, to varying degrees, the meaning of the work. Choice A best captures this viewpoint and is the correct summary of the paragraph.

Option B: Choice B is completely against the viewpoint expressed in the first sentence of the paragraph. The second sentence is very extreme and cannot be validated from the information given in the paragraph.

Option C: Choice C again contradicts the first sentence of the paragraph (all first-rate translators convey the story and spirit of the works at hand). It is also incomplete and does not cover the main points of the paragraph.

Option D: Choice D is wrong as the paragraph does not differentiate between first-rate translators and second-rate translators. Choice D is not an apt summary. Choice (A)

Solution for question 19:

19. The statements immediately preceding and following blank (1) talk about the child finding the project (volunteering to care for a relative with dementia for a couple of hours a day) stressful. The highlighted sentence talks about other aspects like opportunity costs and inserting it between these statements would break the flow. Hence, the highlighted sentence does not fit in blank (1).

"she is therefore unable to assess" in the statement immediately following blank (2) is directly connected to "child's evaluative capacities are not sufficiently mature and developed" in the statement immediately preceding blank (2). Therefore, these statements form a pair and inserting the highlighted sentence between these statements would break the flow. Hence, the highlighted sentence does not fit in blank (2).

"she is therefore unable to assess" in the statement immediately preceding blank (3) is directly connected to "that is why she might end up making…" in the statement immediately following blank (3). The child might end up making mistakes because she is unable to assess such caring obligations. Therefore, these statements form a pair and inserting the highlighted sentence between these statements would break the flow. Hence, the highlighted sentence does not fit in blank (3).

The highlighted sentence adds on to the statement immediately preceding blank (4), which talks about what the child may end up doing because she is unable to assess such projects. The highlighted sentence states that another mistake that the child may end up making is not appreciating that the time caring for relatives takes away precious time doing something more enjoyable and valuable. The statement immediately following blank (4) mentions "such mistakes". Without the highlighted sentence in blank (4), which mentions another mistake, there would only be a single mistake. The other mistakes refer to the ones in the highlighted sentence. Hence, we can infer that "such mistakes" in the statement immediately following blank (4) refers to both the highlighted sentence as well the statement immediately preceding blank (4). Hence, the highlighted sentence fits in blank (4). Choice (D)

Solution for question 20:

20. Sentence 1 introduces the book, giving us the title and what it is about and connects to something else with the expression 'this music.'

Sentence 2 talks about the problems faced by Jazz musicians and how they fought back.

Sentence 3 talks about what 'Jazz' music is and hence, precedes 2, which talks about a sub-idea – problems faced by jazz musicians.

Sentence 4 continues to talk about the evolution of Jazz music with its roots in 'blues'.

The para has two ideas: How Jazz was born, and the social forces which helped it arise and evolve. Sentence 3 introduces the origin of 'jazz' music and will come first. Sentence 4 further provides some explanation of the origin of jazz and will come next. Therefore, 34 is a pair. The introduction of the book, in sentence 1, connects well with 3 and 4, all of which connect jazz music with the social and political issues of the time. Sentence 2 explains more about why the music was called 'phenomenal' in sentence 1. Hence, sentence 2 follows sentence 1. So, 3412 is the block. Ans: (3412)

Solution for question 21:

21. Option A: While the para does exhort the EU to adopt English as its common language, it is not to 'live up' to the EU's cosmopolitan ideal. It is to reaffirm it. The para asserts that adopting English won't be a betrayal of the cosmopolitan ideal. This point is missing in the option. Hence, Option A is not the answer.

Option B: This option is apt as it points out both the ideas: that the EU's cosmopolitan ideal (as indicated by multilingualism to a certain extent) will not be undermined (made smaller) by English. Rather, it would reaffirm the ideal of the EU to support several native tongues. Hence, Option B is the answer.

Option C: The EU needs a common tongue according to the para. However, the para doesn't in any way indicate that only English can bring unity in diversity in Europe. English will complement the diversity and reaffirm the cosmopolitan ideal, the respect for diversity, something not mentioned in this option. Hence, Option C is not the answer.

Option D: The para doesn't put English and machine translation on the same pedestal. English isn't really taking it one step further. English, according to the para, will reaffirm or underline the multilingualism of EU. So, it cannot be said that English will carry forward the advantage brought about by machine translation tools. Hence, Option D is not the answer. Choice (B)

Solution for question 22:

22. On a careful reading of the sentences, it can be observed that sentence 3 is a general sentence that begins the paragraph. It introduces the term: Delusional disorder. Sentence 3 is followed by sentence 2, which elaborates on what the disorder is. "serious mental illness called a "psychosis" in which a person cannot tell what is real from what is imagined" in sentence 3 links with "delusions, unshakable beliefs in something untrue or not based on reality" in sentence 2. Sentence 2 is followed by sentence 4. "presence of delusions" in sentence 2 links with "these delusions" in sentence 4. Sentence 1 (These delusions also involve) follows sentence 4 (These delusions mainly involve ....) and concludes the para. So, 3241. Ans: (3241)

Solution for question 23:

23. The highlighted sentence talks about scientific revolutions occurring in the absence of incentives. "such incentives" needs a precedence. Hence, the highlighted sentence cannot be at the beginning of the paragraph and does not fit in blank (1).

"such incentives" in the highlighted sentence cannot refer to new drugs being cheaper or mitigating financial incentives corrupting medical science, as mentioned in the statements preceding blank (2). Besides, the statement immediately following blank (2) is directly connected to the statement immediately preceding blank (2) (cost of new drugs). Hence, the highlighted sentence does not fit in blank (2).

"such incentives" in the highlighted sentence can refer to defending intellectual property rights. "breakthroughs" in the statement following blank (3) is directly connected to "scientific revolutions"

mentioned in the highlighted sentence. Also, the statement "breakthroughs in medicine are no different" implies that it draws parallels to breakthroughs in something else. These are the scientific revolutions in science. Therefore, this statement and the highlighted sentence are directly connected. Hence, the highlighted sentence fits in blank (3).

The statements immediately preceding and following blank (4) talk about breakthroughs in medicine while the highlighted sentence talks about other scientific breakthroughs. Therefore, inserting the highlighted sentence between them would break the flow. Hence, the highlighted sentence does not fit in blank (4). Choice (C)

Solution for question 24:

24. The para primarily connects education to inequality and explains two solutions. One is the vocational approach while the other is the civic conception of education. The latter is preferred by the author.

Option A: This option separates out the two types of inequality, income and political and mentions that the para offers two different solutions for each. That is not the essence of the para. The para offers civic education as an alternative which in the long run also offers economic solutions. Hence, Option A is not the answer.

Option B: While the first part of this option is true, the second part digresses from the essence of the para by conveying that political awareness of citizens is more effective in reducing income inequality. Political awareness is not exactly similar to what the civic concept of education offers. Even if that approximation is made, the option ignores the idea of political equality that has not been broached in this option. Option B is not the answer.

Option C: This option indicates that focusing on technical skills is not enough as one also needs to be prepared to participate in democratic processes (political equality), which is what the para conveys. Vocational approach is not enough. One needs to be able to participate in political processes in order to also lift oneself out of income inequality. Option C is the answer.

Option D: While this is true to a large part, it skips the important idea of focusing on the civic part of education, instead conveying that political equality is important. It doesn't mention how that can be assured, even though the para does talk about it. Option D is not the answer. Choice (C)

Difficulty level wise summary - Section I

Level of Difficulty	Questions
Very Easy	–
Easy	15, 22, 23
Medium	13, 14, 17, 18
Difficult	1, 3, 5, 9, 11, 16, 20, 21
Very Difficult	2, 4, 6, 7, 8, 10, 12, 19, 24

SECTION – II

Solutions for questions 1 to 5:

Given, from (i), that P1 can be started only after P2, and from (iii), P6, P8, and P3 are not the final process. Also from (v), P4 cannot be the final process. Hence the last two processes must be P7 and P5, in that order, from (ii), and the first two processes must be P2 and P1, in that order.

Arranging this information, we get the possible order as

P2 – P1 – **x** – P6/P8 – **x** – P8/P6 – **x** – P3 – **x** – P7 – P5, where P4 can come at one of the locations indicated by an '**x**'.

The order of machines' locations must be selected such that there are at least eight locations in the order. From the network/figure, it can be seen that this is possible in only two ways:

A → B → G/C → H → I → D → E → J

From (iv), M3 cannot be placed in D. Hence, P3 cannot be the sixth process. This leaves us with only the final possible order as P2 – P1 – P6/P8 – P8/P6 – P3 – **P4** – P7 – P5.

Hence, the machines and the processes are as follows.

A	B	G/C	H	I	D	E	J
P2	P1	P6/P8	P8/P6	P3	P4	P7	P5

1. Location G need not have any machines among the given options (in that case, a machine needs to be at location C). Choice (C)

2. The machines can be placed in 2 ways if a machine is placed on G, and 2 ways if a machine is placed on C. Hence, the machines can be placed in 4 ways in total. Choice (B)

3. The fifth process to occur should be P3. Choice (C)

4. P4 occurs immediately after P3. Choice (D)

5. M6 can be in C or H. Hence, the answer is option A. Choice (A)

Solutions for questions 6 to 10:

For any student, say X, whose marks did not change, the increase (or decrease) in rank can be at most 1. One possible way in which this can happen is as follows. Any student who initially scored 2 marks more than X can have his score decreased by 4. The student who initially scored 4 marks more than X cannot have his score decreased to less than the score of X.

For any student, say X, whose marks increased by 2, the increase in rank can be at most 2. One possible way that this can happen is as follows. The student who initially scored 2 marks more than X can have his score decreased by 2 or 4 marks (in either case his revised score will be less than that of X, since X's score also increased by 2). The student who initially scored 4 marks more than X can also have his score decreased by 4. Hence, X can overtake two students with an increase of 2 marks.

Similarly, for a student, say X, whose marks increased by 4, the increase in rank can be at most 3. Assume that the rank of X is 27 and X's score is a. The students at ranks 24, 25 and 26 can have a score of a + 6, a + 4 and a + 2, respectively. The revised score of X is a + 4.

Now, of the students who have initially scored higher than X, apart from the students at ranks 24, 25, 26, no other students can end up with a score lower than that of X. This is because the student at rank 23 can have a score of at least a + 8. His revised score will be at least a + 4, which is the same as the score of X.

The student whose rank is 24 must have a revised score of a + 2 (with a decrease of 4 marks). Only then can he have a rank worse than that of X. The student whose rank is 25 must have a revised score of a (he cannot have a score of a + 2, since the previous student already has that score). The student whose rank is 26 must have a revised score of a – 2.

Hence, the three students who initially scored 2 marks, 4 marks and 6 marks more than X can have each of their scores reduced by 4 marks. This is the only way in which all the three students will end up with lower marks than X.

From the table, we can see that A's rank increased from 70 to 73. This is possible only if the students whose initial ranks are 71, 72 and 73 had an increase in their score and got revised ranks of 70, 71 and 72, respectively (similar to the explanation above but for an increase in rank).

Also, the rank of F decreased from 27 to 24. This is possible only if the students whose initial ranks are 24, 25 and 26 had a decrease in their score and got revised ranks of 25, 26 and 27 respectively.

Also, given that E's rank increased from 30 to 32 and C's rank increased from 29 to 31. For the rank to increase by 2, their score must change by at least 2.

If C's score decreased by 2, then the student whose initial rank is 30 must have gotten a revised rank of 29 and the student whose initial rank is 31 must have got a revised rank of 30 (with the scores of both of them decreasing by 4). However, it is given that the student whose initial rank is 30 (which is E) has a revised rank of 32. Hence, C's score could not have decreased by 2 and must have decreased by 4.

Let C's initial score be p. The revised score of C must be p – 4. E's initial score can be at most p – 2. His revised score must be at least p – 6 (for E to be ranked worse than C) and at most p – 6 (since his initial score can be at most p – 2). Hence, E's initial score is p – 2 and revised score is p – 6.

The students with a rank numerically higher than 32 cannot have a revised scored more than that of C. Hence, the two students who end up with a higher revised score than C (for C's rank to increase by 2) must be the ones with initial ranks of 31 and 32.

The student whose initial rank is 32 can have an initial score of at most p – 6. His score can increase to p – 2 and can have a revised rank of 30. Since his score must be greater than that of C, the initial score of this student must be p – 6. Similarly, the student whose initial rank is 31 can have an initial score of p – 4 and revised score of p.

Given that the difference between the recalculated marks of B and E is 94. Let the revised score of B be b. The revised score of E must be b – 94. The revised score of C must be b – 92.
From (i), the revised score of A must be b – 92 – 148 = b – 240.

The revised score of B can be at most 296 (since B's revised rank is 3 and the maximum possible score is 300). Hence, the revised score of A can be at most 56.

The initial score of A must be 60. Since the initial rank of A is 70 and there are 100 students in total, there must be 30 students whose score is less than that of A. Only if A's initial score is 60, the possible number of scores less than that of A will be 30. Hence, the initial scores of the students with initial ranks of 71, 72… 100 must be 58, 56, 54 … 0, respectively.

Since the revised score of A must be 56, B's revised score must be 296, E's revised score must be 202 and C's revised score must be 204.

The revised score of F must be at least 218 (since there are six students between C and F according to the revised ranks). Also, the revised score of F can be at most 254 (since there are 20 students between B and F according to the revised ranks). The initial score of F must be at least 214 and at most 250 (since we know that F's score increased by 4). However, since F's initial rank is 27, there must be at least 26 even integers above F's initial score. Hence, F's initial score must be 300 – 26 × 2 = 248.

Since F's initial score is at most 248, his revised score can be at most 252 (with a 4-mark increase). Similarly, we can find the range of marks for the other students.

The following table provides the initial ranks, initial scores, revised ranks and revised scores of the ranks that were discussed above:

Student	Initial Rank	Initial Marks	Revised Rank	Revised Marks
	24	220 – 254	25	216 – 250
	25	218 – 252	26	214 – 248
	26	216 – 250	27	212 – 246
F	27	214 – 248	24	218 – 252
C	29	208	31	204
E	30	206	32	202
	31	204	29	208
	32	202	30	206
A	70	60	73	56
	71	58	70	62
	72	56	71	60

	73	54	72	58

6. The marks scored by A before the recalculation is exactly 60. Choice (A)

7. We know that the revised rank of F is 24 and his revised score is in the range of 218 to 252. A student whose revised rank is 20 will have his revised score as at least 218 + 8 = 226. His revised score will be at most 262 (since there must be 19 students and 38 marks above him). Since the marks of any student can change by at most 4, the lower limit of the initial marks of the student whose revised rank is 20 can be 226 – 4 = 222 and the upper limit of the initial marks of the student whose revised rank is 20 can be 262 + 4 = 266.

Of the given values, except for V, all the other values fall in this range. Hence, 4 values satisfy. Choice (A)

8. The student whose initial rank is 31 has a revised rank of 29. Choice (D)

9. The student whose initial rank is 32 has initial score of 202. Hence, the score of the student whose initial rank is 33 can be at most 200.
Also, the student whose initial rank is 70 (i.e., A) has initial score of 60. Hence, the student whose initial rank is 33 must have a score of at least 60 + (70 – 33) × 2 = 134.
Hence, this student must have marks in the range of 134 to 200.
The student whose initial rank is 50 can have a score of at most 202 – (50 – 33) × 2 = 168.
The minimum possible score for this student is 60 + (70 – 50) × 2 = 100.
Given that these scores are in AP and we need to maximize the difference between the marks of two students. Hence, we need to maximize the common difference of this progression.

We can consider the score of the student whose initial rank is 33 to be 200 and the score of the student whose initial rank is 50 to be 100.
The difference between the scores = 200 – 100 = 100 must be split into 17 parts. However, 100/17 = 5.8823 is not an integer.
We can take the common difference to be 5, but in this case, some of the marks will not be an even integer. Hence, the maximum common difference is 4.
Since we need to find the maximum possible difference between the initial marks of the student whose initial rank is 40 and the student whose initial rank is 42, this must be 4 × 2 = 8 marks.
Ans: (8)

10. Continuing from the above solution, the maximum common difference is 4. If the score of the student whose initial rank is 33 is 200 (which is the maximum possible), then the score of the student whose initial rank is 50 will be 200 – (50 – 33) × 4 = 132.
The score of the student whose initial rank is 55 can be at most 132 – (55 – 50) × 2 = 122.
Ans: (122)

Solutions for questions 11 to 15:

above (if he is below the fifth floor, he cannot share the lift for five floors). Also, from (ii), Anurag lives immediately above Dev. This implies that Dev cannot be on the 8th floor. From (vii), Chintu lives above Anurag. Hence, Dev cannot be on the 7th floor and Anurag cannot be on the 8th floor. Therefore, Dev can be on the 5th or 6th floor.

If Dev is on the 5th floor, Anurag will be on the 6th floor. From (v), the person on the third-floor leaves for office at 10:00 AM. But from (iv), the person who lives three floors below Anurag has to leave for office at 9:00 AM. Hence, Anurag cannot be on the 6th floor and Dev cannot be on the 5th floor.

Hence, Dev has to be on the 6th floor and Anurag has to be on the 7th floor. Chintu will then be in the 8th floor. Since no person shared the lift for more than five floors, the persons living on the 6th, 7th and 8th floors must leave at three different times. Since Dev leaves at 8:00 AM (from (vii), he shares the lift with a person who leaves at 8:00 AM), Anurag can leave at 9:00 AM or 10:00 AM. From (ii), Anurag can leave only at 10:00 AM and Chintu will leave at 9:00 AM.

Since Dev has to share the lift with someone for five floors, the person on the fifth floor should also leave at 8:00 AM. In this case, Dev can share the lift for five

11. At no point of time in the movie were there exactly five actors on screen. Choice (A)

12. 10 actors appeared on screen between 00:20 and 00:25. Ans: (10)

13. There were more than eight actors on screen between 00:20 to 00:35; 00:45 to 00:50; 00:55 to 01:00; 01:05 to 01:10, i.e., for a total of 30 minutes. Ans: (30)

14. J will be the first name to be displayed. D, F and M will be the 2nd, 3rd and 4th names to be displayed in some order. A, K and O will be the 5th, 6th and 7th names to be displayed in some order. B and N will be the 8th and 9th names to be displayed. Between B and N, N appeared for 65 minutes, while B appeared for 60 minutes. Hence, N will be the 8th name and B will be the 9th name to be displayed. Choice (B)

15. J is the first name and D, F and M are the 2nd, 3rd and 4th names, in some order. A, K and O are the 5th, 6th and 7th name, in some order. Since we need the names of the last nine actors, we need to identify who among A, K and O were 5th, 6th and 7th. A appeared for 40 minutes, K for 25 minutes and O for 35 minutes. Hence, A would be the 5th name, O and K would be the 6th and 7th names respectively.
Ignoring J, D, F, M, A and O, among the remaining actors, N appeared for 65 minutes, which is the longest. Ans: (65)

Solutions for questions 16 to 20:

From (vii), Dev shares the lift with someone else for five floors. Hence, Dev has to be on the fifth floor or floors with the person living on the fifth floor.

From (vi), the person on the fourth-floor leaves at 9:00 AM. Pranav is on the second floor and he also leaves at 9:00 AM. Chintu, Pranav and the person on the fourth floor will be the only persons leaving at 9:00 AM (from (i)). From (vii), the person on the fourth floor has to be Bhim.

From (v), the person on the third-floor leaves at 10:00 AM. This cannot be Satish or Varun (from (v) and (iii)). Hence, this has to be Rajiv. The person on the first floor and the fifth floor can be Satish and Varun in any order. Also, the person on the first floor cannot leave at 10:00 AM in either case and can only leave at 8:00 AM (since neither Satish nor Varun leaves at 10:00 AM).

The following table provides the arrangement:

Floor	Person	Time
8	Chintu	9:00 AM
7	Anurag	10:00 AM
6	Dev	8:00 AM
5	Satish/Varun	8:00 AM
4	Bhim	9:00 AM
3	Rajiv	10:00 AM
2	Pranav	9:00 AM
1	Varun/Satish	8:00 AM

16. Bhim lives on the fourth floor of the building. Choice (C)

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17. Rajiv will share the lift only with Anurag. The remaining share a lift with two persons. Choice (C)

18. Chintu, who lives on the eighth-floor, leaves for office at 9:00 AM. Choice (B)

19. Anurag lives above Rajiv and leaves at the same time as Rajiv. Choice (B)

3. The sum of the first n terms of the first series (i.e., S_1) = $\frac{n}{2}[2+(n-1)1]$

Similarly $S_2 = \frac{n}{2}[8+(n-1)2]$,

$$S_3 = \frac{n}{2}[18+(n-1)3]$$

$\therefore S_1 + S_2 + S_3 + \ldots\ldots\ldots\ldots + S_m$

20. Even with the given information, either Varun or Satish can be on the first floor. Hence, the answer is either Satish or Varun. Choice (D)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	11, 12, 13, 14, 15
Medium	1, 2, 3, 4, 5
Difficult	16, 17, 18, 19, 20
Very Difficult	6, 7, 8, 9, 10

SECTION – III

Solutions for questions 1 to 5:

1. We can take the sum in
$3 \times 3 \times 3 = 27$ ways.
$\therefore$ The sums must be each integer from 1 to 27.
As we are looking for the least possible value of the greatest integer, that must be as close as $\frac{27}{3} = 9$.
Let a, b and c be the greatest integers in each group, with a > b > c.
$a + b + c = 27$.
As a = b = c = 9 is not possible, a is at least equal to 10.
Take a = 10, b = 9 and c = 8.
We get sum values as all the integers from 1 to 27, only if the other integers in the first, second and the third groups are (a – k, a – 2k), (b – l, b – 2l), (c – m, c – 2m) respectively.
When (k, l, m) = (9, 3, 1) in any order.
If k = 1, a – k = 9, which is not possible as b = 9.
If k = 2, a – k = 8, which is not possible as c = 8.
If k = 3, a – k = 7.
But l = 2 or m = 1 $\Rightarrow$ either b – 2l = 7 or c – l = 7.
$\therefore$ Take a = 11, b = 10, c = 6.
If k = 9, l = 3 and m = 1, the given all conditions will be satisfied.
$\therefore$ The required answer is 11.
Ans: (11)

2. The sum of all the exterior angles of any convex polygon is 360°.
Let the number of sides of the polygon be n
Sum of all the interior angles = $(n–2)\ 180°$
It is given that $(n–2)\ 180° = 7\ (360°)$
$\Rightarrow n – 2 = 14$
$\Rightarrow n = 16$. Ans: (16)

$$- \frac{n}{2}\left[\left(2+8+18+\ldots\ldots+2m^2\right) + (n-1)(1+2+3+\ldots\ldots\ldots\ldots m)\right]$$

$$= \frac{n}{2}\left[2\left(1+4+9+\ldots\ldots+m^2\right) + (n-1)(1+2+3+\ldots\ldots\ldots\ldots m)\right]$$

$$= \frac{n}{2}\left[\frac{m(m+1)\ (2m+1)}{3} + (n-1)\frac{(m)(m+1)}{2}\right]$$

$$= \frac{n}{2}\ [m(m+1)]\left[\frac{2m+1}{3} + \frac{n-1}{2}\right]$$

$$= \frac{mn(m+1)}{2}\left(\frac{4m+3n-1}{6}\right)$$

Alternate Solution:

We can substitute values and validate the options. Taking m = n = 1 will eliminate Choice C. Taking m = 2 and n = 1 will eliminate Choices A and B. Choice D will be the answer. Choice (D)

4. Among the conclusions, 13 ordered pairs appear but (a ∗ b) depends only on the product ab.
$\therefore (3 * 2) = (2 * 3) = (1 * 6) = (6 * 1) = – 4$
$(4 * 2) = (8 * 1) = – 6$
$(–1 * 2) = (–2 * 1) = 4$
$(2 * 8) = – 14$
$6 * 3 = – 16,\ 1 * 2 = 0,\ 7 * 3 = – 19$
We consider the conclusions
I. Given equation = $(– 4) * (– 4) = –14$ - True
II. LHS = $(– 6) * (– 4) = – 22$
RHS = $(– 4) * (– 6) = – 22$ - True
III. LHS = $4 * 4 = –14$
RHS = – 14 - True
IV. LHS = $(– 4) * (– 4) = –14$
RHS = $(– 19) * 0 = 2$ - False
$\therefore$ Exactly three statements are true. Ans: (3)

5. Since the job is to be done in minimum number of days, P should work for the maximum possible number of days. So, we should group them as PQ and PR so that 'P' will be working right from the first day to the last day.
PQ take 60/9 days. PR take 60/8 days
Since PQ are faster, PQ start the work, work for one day and leave and then PR work for one day and leave and they continue in this fashion.
$\therefore$ Amount of work done in 2 days = $\frac{9}{60} + \frac{8}{60} = \frac{17}{60}$.
In 6 days 51/60 of the work is over. On the 7th day the remaining 9/60 of the work is completed.
$\therefore$ The work is done in 7 days. Choice (D)

Solutions for questions 6 and 7:

Let the total quantities of the water in the water drum and the milk in the milk drum be W and M respectively.

Let the concentration of milk that Bhola got be, say C_1.

Now $C_1 = \dfrac{\frac{M}{2}}{\frac{M}{2}+W} = \dfrac{M}{M+2W}$

Let C_2, be the concentration of milk in the water drum if Bhola had done as Madhu asked him to (i.e., poured into it half of the water from the water drum)

$\therefore C_2 = \dfrac{M}{\left(M+\frac{W}{2}\right)} = \dfrac{2M}{2M+W}$

Given $\dfrac{C_1}{C_2} = \dfrac{1}{n} = \left(\dfrac{M}{M+2W}\right) / \left(\dfrac{2M}{2M+W}\right)$ ----- (1)

$\Rightarrow \dfrac{M}{W} = \dfrac{4-n}{2n-2}$ (by simplifying equation (1)) ----- (2)

6. Given $n = 3$

   $\Rightarrow \dfrac{M}{W} = \dfrac{4-3}{2\times3-2} = \dfrac{1}{4}$

   $\Rightarrow W = 4M$

   If Madhu mixes W and M, then $\dfrac{M}{M+W} = \dfrac{M}{M+4M}$

   = 20% Choice (C)

7. From equation (1),
   If M = 4.5W,

   Then $n = \dfrac{4W+2M}{2M+W} = \dfrac{13W}{10W} = 1.3$ Choice (A)

Solutions for questions 8 to 22:

8. Any number of the form K^2 (where $K \in N$) has an equal number of factors more than K and less than K.
   For instance, factors of $2^2 = 4$ are 1, 2, 4. There is one factor more than 2 and one factor less than 2.
   $\therefore$ If N has 7 factors more than $\sqrt{N}$, then it must have 7 factors than $\sqrt{N}$.
   $\therefore$ It has a total of 7 + 7 + 1 = 15 factors (including $\sqrt{N}$)
   $\therefore$ N can be of the form a^2b^4 or a^{14} ($\because 15 = 3 \times 5$ OR 1×15)
   $\Rightarrow \sqrt{N} = ab^2$ or a^7
   $\Rightarrow N^{3/2} = a^3b^6$ or a^{21}
   $\therefore$ It has (3 + 1) (6 + 1) = 28 OR 21 + 1 = 22 factors.
   $\therefore$ It can have at most 28 factors.

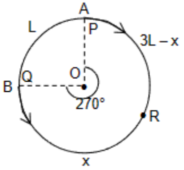
   Ans: (28)

9. Let the population of A be 100
   Therefore, the population of B will be 300.

	A	B	Total
No. of octogenarians	2	18	20

Therefore, the percentage of octogenarians = $\dfrac{20}{400} \times 100 = 5\%$ Choice (B)

10. Let the total length of the track be 4L m



Let x m be the distance covered by B by the first meeting at R
$\Rightarrow$ A covers (3L – x)m by the first meeting
As A and B should cover 4L for the second meeting instead of 3L, A has to cover $\dfrac{4}{3}$ times the distance covered by him for the first meeting.

Given $x = \dfrac{4}{3}(3L - x)$

$\Rightarrow 3x = 12L - 4x$

$\Rightarrow x = \dfrac{12}{7}L$ and $3L - x = \dfrac{9L}{7}$

Hence, the fraction of the total length covered by A by the first meeting $= \dfrac{\left(\frac{9L}{7}\right)}{4L} = \dfrac{9}{28}$ Choice (B)

11. Given $f(x) = \dfrac{x}{1+x^2}$

The domain of $f(x)$ is R.

$g(x) = \dfrac{e^{-x}}{1+\lfloor x \rfloor}$

Now $1+[x]$ should not be 0

$\Rightarrow x \notin [-1, 0)$

$\therefore$ Domain of $g(x)$ is R – $[-1, 0)$

$\therefore$ Domain of $f(x) + g(x)$ = R – [1, 0)
We shall try to find the range of $f(x)$

Let $y = \dfrac{x}{1+x^2} \Rightarrow y + x^2y = x$

$\Rightarrow x^2y - x + y = 0$

$\Rightarrow x = \dfrac{1 \pm \sqrt{1-4y^2}}{2y}$

Clearly y cannot be 0, and $\sqrt{1-4y^2} \geq 0$

Thus $y \neq 0$ and $1 - 4y^2 \geq 0$

$\therefore y \neq 0$ and $y \in \left[-\dfrac{1}{2}, \dfrac{1}{2}\right]$

$\therefore$ Range of $f(x) = \left[-\dfrac{1}{2}, \dfrac{1}{2}\right] - \{0\}$

Consider $g(x) = \frac{e^{-x}}{1+\lfloor x \rfloor}$

Let $y = \frac{e^{-x}}{1+\lfloor x \rfloor}$

$y > 0$ if $[x] > -1$

And $y < 0$ if $[x] < -1$

$\therefore$ Range of $g(x) = R - \{0\}$

Hence, range of $f(x) \cap$ range of $g(x)$

$= \left[-\frac{1}{2}, \frac{1}{2}\right] - \{0\}$

Thus, statements II and IV are true. Choice (A)

12. Using statement I alone: $|y| > 1$

$\Rightarrow y \in (-\infty, -1) \cup (1, \infty)$

for which, $|y - 2|$ can be greater than or less than 1.

Hence, statement I alone is not sufficient.

Using statement II alone: $|y - 1| < 2$

$\Rightarrow y \in (-1, 3)$

for which, $|y - 2|$ can be greater than or less than 1.

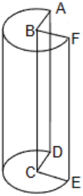
Hence, statement II alone is not sufficient.

Using both statements I and II, we get that

$y \in (1, 3)$

for which, $|y - 2|$ is always less than one, i.e., though the exact value of y is not known, the answer to the question "Is $|y - 2| < 1$?" can be determined as "Yes". Choice (C)

13.



The sum of the surface areas of BFEC and BCDA (which are rectangles) = 2 × BF × BC = 2 × 14 × 20 = 560 cm². Total area of the flat surfaces

$= 2\,[3\pi/4 \times (14)^2]$

$2 \times 3/4 \times 22/7 \times 14 \times 14 = 924$ cm²

Area of the curved surface $= 3/4 \times 2\pi \times 14 \times 20$.

$= 3/4 \times 2 \times 22/7 \times 14 \times 20 = 1320$ cm²

So total surface area of the new block

= (560 + 924 + 1320) = 2804 cm² Choice (D)

Substituting $\frac{4}{a}$ in $x^2 - ax + 3 = 0$, we get

$\frac{16}{a^2} - 4 + 3 = 0$

$\Rightarrow a = \pm 4$

Only for $a = 4$, we get a positive root (i.e., 1) in common.

Hence, the value of a is 4. Ans: (4)

15. Since the age of Y is 4 years less than that of X, the average age of the 11 players after player X is replaced by player Y will become $\frac{34.11 - 4}{11}$

$= 34 - \frac{4}{11}$.

After 1 more year, the average age of the new team of 11 players will become $35 - \frac{4}{11} = 34\frac{7}{11}$ years.

Now, it is given that, when another player, Y, is replaced by a new player, Z, the average becomes $34\frac{4}{11}$ years.

Hence, the age of the player Z is $\left(34\frac{7}{11} \times 11 - 34\frac{4}{11} \times 11\right)$, i.e., 3 years less than that of player Y. Choice (A)

16. Let C_1, C_2, C_3, C_4, C_5, C_6, C_7, C_8 and C_9 be the corners of the 9 – side regular polygon. The favourable combinations of forming a quadrilateral are:

(C_1, C_3, C_5, C_7), (C_1, C_3, C_5, C_8), (C_1, C_3, C_6, C_8), (C_1, C_4, C_6, C_8), (C_2, C_4, C_6, C_8), (C_2, C_4, C_6, C_9), (C_2, C_4, C_7, C_9), (C_2, C_5, C_7, C_9) and (C_3, C_5, C_7, C_9).

Joining the four corners in each of these combinations we get a desired quadrilateral. Hence, the required number of quadrilaterals is 9.

Alternative solution:

(i) With exactly one side as common to quadrilateral and nonagon.

$\Rightarrow 9 \times ({}^5C_2 - 4) = 54$ (9 ways of choosing common side x ${}^5C_2$ ways of choosing other two points of which 4 ways yield two common sides)

14. Let the root common to both the equations be denoted by t.

$\therefore t^2 - at + 3 = 0$ and $t^2 + at - 5 = 0$

$\therefore t^2 - at + 3 = t^2 + at - 5$

$\therefore 2at = 8$

$t = \frac{4}{a}$

(ii) With exactly two sides in common $9 \times 4 + \frac{9 \times 4}{2}$

$= 54$ (9 ways of choosing two adjacent sides x four ways of choosing a fourth vertex + nine ways of choosing one side and 4 ways of choosing opposite side)

(iii) With exactly three sides as common → there are nine possible quadrilaterals.

(iv) With exactly four sides as common → zero possible quadrilaterals.

⇒ With no side as common = total quadrilaterals – quadrilaterals with exactly 1 or 2 or 3 or 4 sides.

$= {}^9C_4 - 54 - 54 - 9 = 126 - 117 = 9.$

Choice (C)

17. Let us say Rohan had ₹x at the beginning of the first round. The amounts with him at the beginning and end of each of the first 3 rounds would be

Round No.	Amount at beginning (in ₹)	Amount at end (in ₹)
1	x	$2x - 2048$
2	$2x - 2048$	$2^2x - (2^2 - 1)\,2048$
3	$2^2x - (2^2 - 1)\,2048$	$2^3x - (2^3 - 1)\,2048$

It can be seen that at the end of the n^{th} round he was left with ₹$[2^n x - (2^n - 1)\,2048]$

When $n = 10$, he would be left with

₹$[2^{10}x - (2^{10} - 1)\,2048]$ = ₹0 (given)

$x = 2046$

Note: If he started with an amount more than ₹2048, the amount with him would keep on increasing. If he started with ₹2048, he would have ₹2048 at the end of each round.

∴ For him to be left with no money at the end of 10 rounds, he must start with an amount less than ₹2048. This observation alone can eliminate both choices (A) and (B). Choice (C)

18. Let the salaries of Raghu, Ram and Rajan be $2k$, $3k$ and $5k$ respectively.

Their total salary $= 2k + 3k + 5k = 10\,k$.

It is given that, $10k = 720000$

$\therefore 2k = 144000$.

Therefore, Raghu's monthly salary is ₹1,44,000

Ans: (144000)

all of them slightly downwards, the lengths of C_1, C_2, C_3, C_4 will increase and the lengths of C_6, C_7, C_8, C_9 (as well as C_5) will decrease. But the decrease will not be equal to the corresponding increases. Therefore, it is not necessary that any two of the chord should be of equal length.

III. The difference between C_3 and C_4 will be less than 1 cm

Therefore, none of the given statements is true.

Choice (D)

20. Given $a_1 = h_1 = 4$; $a_{10} = h_{10} = 5$

Let 'd' be the common difference of the A.P.

$a_{10} = a_1 + 9d$

$5 = 4 + 9d$

$\frac{1}{9} = d$

$a_4 = a_1 + 3d = 4 + 3 \times \frac{1}{9} = \frac{13}{3}$

There is also a harmonic progression given, where $h_1 = 4$ and $h_{10} = 15$.

$\Rightarrow \frac{1}{h_1} = \frac{1}{4}$; $\frac{1}{h_{10}} = \frac{1}{5}$ and the series is in A.P.

Let d_1 be the common difference of second A.P.

$\frac{1}{h_{10}} = \frac{1}{h_1} + 9d_1$

$\frac{1}{5} = \frac{1}{4} + 9d_1$

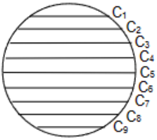
$d_1 = \frac{\frac{1}{5} - \frac{1}{4}}{9} = -\frac{1}{180}$

$\therefore \frac{1}{h_7} = \frac{1}{h_1} + 6d_1 = \frac{1}{4} - 6\left(\frac{1}{180}\right) = \frac{15-2}{60} = \frac{13}{60}$

$\therefore h_7 = \frac{60}{13}$

Therefore, $a_4 h_7 = \frac{13}{3} \times \left(\frac{60}{13}\right) = 20$ Ans: (20)

19. We consider the options



I. It is possible to have 9 chords as shown above, where C_5 is a diameter. But we can displace all the chords by a small amount (say 0.1 cm) and none of the chords is a diameter.

II. Again, in the initial set of chord $C_1 = C_9$, $C_2 = C_8$, $C_3 = C_7$, $C_4 = C_6$. But if we displace

21. Let the two digits of the number be $\underline{A}\,\underline{B}$

$\therefore$ Value of the number is 10A + B

Given

$(10A + B) - (A + B) = X(A + B)$

$9A = XA + XB$

$A(9 - X) = XB \Rightarrow XB + (X - 9)A = 0$ ---- (1)

Now when the digits of the number are interchanged the value of the number becomes 10 B + A.

Difference between the number and the sum of it, digits $= 10B + A - A - B = 9B$

$= 9B - XB - (X - 9)A$ [$\because$ $XB + (X - 9)A = 0$]

$= (9 - X)(A + B)$

$= (9 - X)$ times the sum of its digits.

Choice (A)

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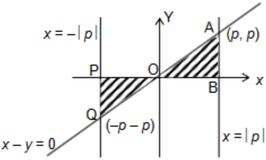
22. The equation $|x| = |p|$, will be two parallel lines $x = |p|$ and $x = -|p|$

(The condition that $|p| \geq 1$ is not important)

The line $x = y$ will intersect as shown in figure.



Are required = Area of triangle OPQ + Area of triangle OAB

Since triangle OPQ, OAB are equal in Area.

Area required = 2(Area of OAB) $= 2\left(\frac{1}{2}\right)(p)(p)$

$= p^2$

Choice (B)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	18
Easy	2, 4, 9, 13, 14
Medium	3, 5, 10, 12, 15, 17, 19, 21, 22
Difficult	1, 6, 7, 8, 11, 16, 20
Very Difficult	–